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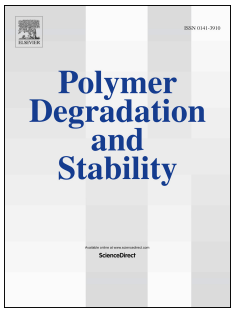

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PII: S0141-3910(16)30396-2

DOI: 10.1016/j.polymdegradstab.2016.12.017

Reference: PDST 8138

To appear in: *Polymer Degradation and Stability*

Received Date: 28 September 2016

Revised Date: 11 December 2016

Accepted Date: 28 December 2016

Please cite this article as: Wang W, Meng L, Leng K, Huang Y, Hydrolysis of waste monomer casting nylon catalyzed by solid acids, *Polymer Degradation and Stability* (2017), doi: 10.1016/ j.polymdegradstab.2016.12.017.

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Hydrolysis of waste monomer casting nylon catalyzed by solid acids

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Abstract

Decomposing the waste polyamide material onto economically-valuable monomer under the relatively mild hydrothermal conditions is a key technology for the development of wastes recycling. This reaction is traditionally catalyzed by homogeneous acids which would result in the difficult separation and equipment corrosion. We first indicated the solid catalysts for the hydrolysis of waste monomer casting nylon in subcritical water, including a series of γ-Al_2O_3 supported solid acid catalysts and some commercial zeolites. Zeolite Hβ-25 exhibited the highest activity among the H-form zeolites, and a better recyclability than the γ-Al_2O_3 supported solid acids. According to the kinetic analyses and reaction pathway exploration, the generation and consumption of linear oligomers, which are the intermediate products, were accelerated when using the zeolite Hβ-25 because of the microporous structure.

Keywords: Catalyzed hydrolysis; Solid acids; Waste nylon; Reaction kinetics

1 Introduction

As one of the most-produced thermoplastic polymers, polyamide 6 (PA 6) is widely used in the synthetic fibers as well as the engineering plastics. The recycling of waste polyamides has always been attracting extensive attentions to minimizing the resource waste and environmental pollution [1]. Compared to the physical recovery method, the monomer recycling is a more potentially promising waste-management method from the environmental point of view [2, 3]. The monomer ε-caprolactam(CPL), which is the precursor to polyamide 6, is generally agreed to be the principal chemical degradation product. Pyrolysis and hydrolysis are two most commonly used decomposition ways to recover CPL, except that the energy cost of pyrolysis degradation is relatively high [4-6].

Consequently, the hydrolysis method, traditionally catalyzed by liquid acids, is more environment-friendly and competitive for the degradation of PA 6 (nylon 6) [7-9]. Considering the difficulty in the product separation, the formation of some acid wastes and the corrosion problems for the experimental device, the traditional homogeneous acids catalyst might not be an optimal selection for the degradation[10-13].

The development of some efficient, easily separable and environmentally friendly heterogeneous catalysts is therefore stimulating interests and considered for the hydrolysis of bio-derived compounds and polymers [11-16]. Faba et al. [17] used different beta zeolites as catalysts in the hydrolysis of hemicellulose. The application of bifunctional zeolite catalysts converted the hemicellulose into valuable polyols

(galactitol, arabitol) due to the presence of acid sites, which could enhance the hydrolysis onto sugars. The hierarchical H-USY zeolite prepared by oxalic acid treatment was also confirmed to be an effective catalyst for the hydrolysis of hemicellulose [18] which could increase the yield of total reducing sugars (TRS) remarkably. Masaru Watanabe et al. [19] conducted the batch experiments for hydrogen production from biomass (glucose and cellulose) with ZrO_2 catalyst in supercritical water (400-440 ℃ and 30–35 MPa) and the yield was almost twice as compared with that without the catalyst. Xue-Kun Li et al. [20] reported that in supercritical CO_2, the waste PET could be almost completely hydrolyzed onto the monomers of terephthalic acid, ethylene glycol and a little amount of diethylene glycol after 12 h with the solid acid catalyst SO_4^{2-}/TiO_2, which could provide H_3O^+ by contacting with H_2O and accelerate the rate of hydrolysis.

Although some progress has been made in the hydrolysis reaction with heterogeneous catalysts, little attention has been devoted to the degradation of waste polyamide materials using solid catalysts. Since supercritical fluids [21, 22], especially the sub- and supercritical water, have been found to be of an increasingly widespread utilization in the green chemistry [23], the degradation of polyamide 6 in supercritical water was also studied. The result showed that the yield of monomer CPL depended on the hydrolysis temperature and reaction time [24]. In view of the harsh condition of supercritical water (374℃, 22.1 MPa), herein we report an efficient hydrolysis of the monomer casting nylon (MC nylon) in relatively mild subcritical water. As a particular kind of polyamide 6 [25], the potential energy requirement for

the conversion of MC nylon into monomer CPL could be decreased by using the solid acid catalysts. A series of γ-Al_2O_3 supported solid acid catalysts and some commercial zeolites were employed to promote the hydrolysis reaction. We found that the hydrolytic degradation processes using various catalysts are different due to the different surface structure of the solid acids. Furthermore, the efficiency of the solid catalyst is more evident at low decomposition temperature which has the potential to reduce the energy consumption with the premise of increasing the monomer yield.

2. Experimental

2.1 Materials

The ivory MC nylon sample for this study was unmodified and collected as processing wastes. Before hydrolysis, the sample was crushed into small pieces (1-7 mm^2), cleaned by water, dried and preserved in a desiccator for use. Chromatographically pure ε-caprolactam was obtained from J & K (Beijing, China). Other reagents and chemicals were of analytical grade from Sinopharm Chemical Reagent Company (Shanghai, China). The water used in hydrolysis experiments was distilled water.

2.2 Preparation of the solid acid catalysts

The commercially available zeolites used throughout the study were provided by Nankai University Catalyst Co. (China) without binder material contents, which were calcined at 550 ℃ in air for 5 h prior to use. H-form zeolites in the protonic form including H-Beta zeolites with the SiO_2/Al_2O_3 ratio of 25 and 40, HUSY with 5 and

HZSM-5 zeolites with 25, 40 and 80 were referred as Hβ-25, Hβ-40, HUSY, HZSM-5-25, HZSM-5-40, and HZSM-5-80 in this work respectively. The regeneration of zeolites catalysts began with drying the recovered catalysts from the recycle experiments at 80 ℃ for 6 hours, following by the calcination in air at 550 ℃ for 5 hours.

The γ-Al_2O_3 (referred as catalyst A) was prepared by calcining the pseudo-boehmite purchased in Nankai University Catalyst Co. (China) at 550 ℃ in air for 5 h. To prepare the SO_4^{2-}/γ-Al_2O_3 and $S_2O_8^{2-}$/γ-Al_2O_3 (labeled as AS1 and AS2, respectively), γ-Al_2O_3 was impregnated in H_2SO_4 (0.5 mol/L) and $(NH_4)_2S_2O_8$ (0.25 mol/L) respectively. The solution/solid ratio was 10 mL/g and the mixture was stirred at 500 rpm for 1 h. After filtered through the filter paper, the solid was subsequently dried at 110 ℃ for 6 h and calcined at 600 ℃ for 3 h to obtain AS1 and AS2. The γ-Al_2O_3 supported ZrO_2 catalyst named AZ was prepared by a precipitation method. An appropriate amount of $ZrOCl_2 \cdot 8H_2O$ was dissolved in water, and then γ-Al_2O_3 was added to this aqueous solution with the molar ratio 30% of Zr/Al. After adjusting the pH value to 9 by aqueous ammonia, the mixture was aged for 24 h. The precipitate solid was filtered, washed thoroughly with de-ionized water until the chloride ion in the filtrate could not be identified. The γ-Al_2O_3/ZrO_2 solid was then dried at 110 ℃ for 6 h and calcined at 600 ℃ for 3 h to obtain AZ catalyst. The SO_4^{2-}/γ-Al_2O_3/ZrO_2 and $S_2O_8^{2-}$/γ-Al_2O_3/ZrO_2 named as AZS1 and AZS2 respectively were prepared by the similar method as AS1 and AS2, but using the γ-Al_2O_3/ZrO_2 solid instead of γ-Al_2O_3.

2.3 Catalyzed hydrolysis in subcritical water

The catalyzed/ non-catalyzed hydrolysis of MC nylon was performed in a 90 cm^3 custom-made high-pressure vessel, which was made of stainless steel and equipped with a pressure gauge (Supplementary Fig. S1). After adding 1.0 g MC nylon sample and desired amount of catalyst in water (mass ratio of liquid /solid was 30:1), the autoclave was sealed. The auto generous pressure in the reactor with added water began to grow about 3 min after putting the sealed autoclave into the salt bath (mass ratio mixture of 0.45:0.55 $NaNO_2$ and KNO_3) and this time was defined to be the start. The system pressure at different temperatures is listed in Table S1.

The autoclave was cooled to room temperature quickly by ice water after the specified time at desired conditions. After filtering and washing, the solid residues, which were a mixture of residual MC nylon and solid catalyst, was dried at 80 ℃ for 6 hours. The liquid product was diluted and then filtered with 0.45 μm filter membrane for further analysis. The ratio of the MC nylon mass at time *t* ($m_{t,nylon}$) to initial mass ($m_{0,nylon}$) is defined as R_t, which is shown in Eq. (1),

$$R_t = \frac{m_{t,nylon}}{m_{0,nylon}} \times 100\% = \frac{m_{t,solid} - m_{catalyst}}{m_{0,nylon}} \times 100\% \quad (1)$$

where $m_{t,nylon}$ is calculated by the difference of the total residue solid mass at time *t* ($m_{t,solid}$) and the mass of solid catalyst ($m_{catalyst}$). For the catalyst recycling experiments, the solid residues after washed and dried can be reused directly for the next recycle. The required amount of new MC nylon feedstock ($m_{new,nylon}$), which was determined by the R_t value of the previous recycle, is calculated by Eq. (2).

$$m_{new,nylon} = m_{0,nylon} \times (1 - R_t) \quad (2)$$

The degree of MC nylon hydrolysis (D_h), the yield of relevant monomer

ε-caprolactam (Y_{CPL}) and oligomers ($Y_{Oligomers}$) in the liquid phase of hydrolysis products are defined as:

$$D_h(\%) = \frac{m_{0,nylon} + m_{catalyst} - m_{t,solid}}{m_{0,nylon}} \times 100 \quad (3)$$

$$Y_{CPL}(\%) = \frac{m_{t,CPL}}{m_{0,nylon}} \times 100 \quad (4)$$

$$Y_{Oligomers}(\%) = \frac{m_{0,nylon} - m_{t,nylon} - m_{t,CPL}}{m_{0,nylon}} \times 100 \quad (5)$$

where $m_{t,CPL}$ referred to the measured mass of CPL at time *t*.

2.4 Analyses and characterization

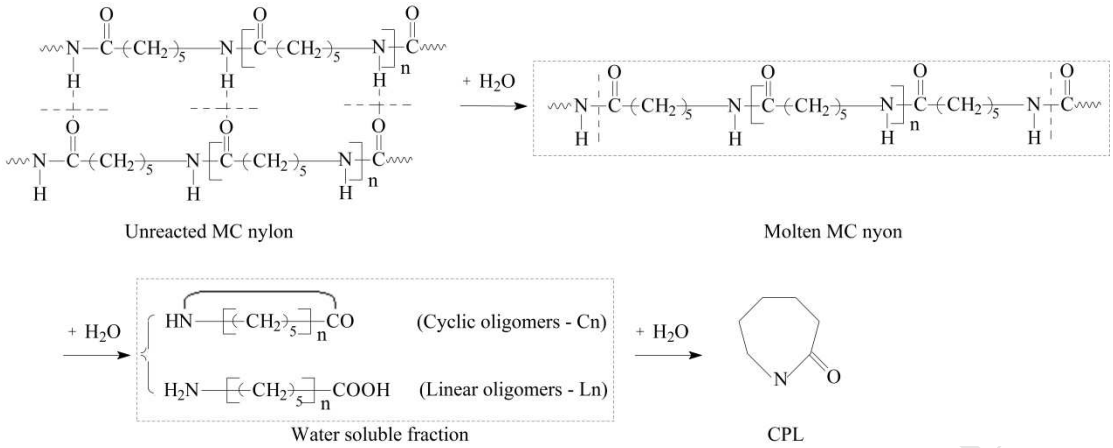
X-ray diffraction (XRD; Empyrean) with the scan range of 2θ from 10° to 80° was employed to confirm the crystal structure of the self-made solid catalysts mentioned before. The surface areas of the solid catalysts were calculated by the BET method, and the nitrogen adsorption–desorption isotherms were measured at -196 ℃ (3H-2000PSI). The temperature-programed desorption of NH_3 (NH_3-TPD) measurements was carried out on FINESORB-3010 with a thermal conductivity detector. After removing the adsorbed species and physically adsorbed NH_3, the TPD profile was determined by increasing temperature from 30 ℃ to 600 ℃ at a heating rate of 5 ℃/min. The contents of the carbon and sulfur elements were estimated by elemental analysis (VARIO Micro cube, ELEMENTAL). Scanning electron microscopy (SEM; Quanta 200FEG, Hitachi Instrument, Inc., Japan) was employed to observe the surface morphology of the solid catalysts and the incompletely hydrolyzed MC nylon.

To analyze the catalyzed hydrolysis process of MC nylon in subcritical water, the

liquid products collected at different reaction time were examined. Liquid chromatography mass spectrometry (LC-MS) and high performance liquid chromatography (HPLC) with gradient elution method were used to analyze the liquid phase products. The mobile phase was water: acetonitrile = 90:10. The LC-MS instrument was Hybrid Finnigan LCQ DECA XP MAX system with ASB-C18 chromatographic column (5 μm, 150 mm×4.6 mm). The m/z range was 50-1500; the flow rate was 0.3 mL/min; the sample size was 5 μL. The HPLC apparatus (Agilent 1100) used the chromatographic column Zorbax SBC-18 (5 μm, 150 mm×4.6 mm); UV detector read at 215 nm; the temperature of the column was 30 ℃.

3. Results and discussion

As a special kind of nylon 6, the primary product of MC nylon hydrolysis reaction is the corresponding synthetic monomer ε-caprolactam (CPL). In subcritical water, the hydrogen bonds between the MC nylon molecules were cleaved under the hydrothermal effect, followed by the melting process because the reaction temperature was much higher than the melting point of MC nylon (219 ℃). Then the chain scission, which was caused by the nucleophilic substitution, generated the water-soluble oligomers and monomer without any gas phase products or harmful by-products as shown in Scheme 1. At the same time, the particle size of incompletely hydrolyzed MC nylon decreased gradually as the degree of hydrolysis increased till the decomposition completed (Fig.1). Our previous showed that the monomer CPL was produced by the cyclization and dehydration of the cyclic and linear oligomers (which were referred as Cn and Ln respectively) step by step in the liquid phase[7]. While the hydrolysis of polyamide is catalyzed by liquid acids traditionally, the effect of solid acids in catalyzed degradation of waste nylon needs further discussion.



Scheme1 Main reaction pathway of MC nylon hydrolysis in subcritical water.

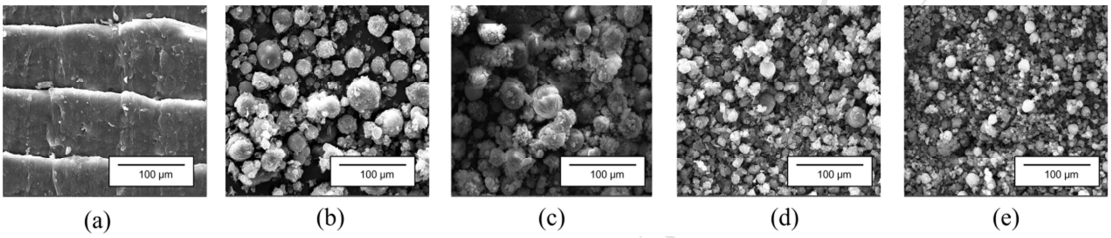


Figure 1. SEM characterization of the surface morphology of (a) Raw MC nylon, incompletely hydrolyzed MC nylon obtained at (b) 10 min, (c) 20 min, (d) 30 min and (e) 40 min from the non-catalyzed hydrolysis reaction system (Temperature: 345 ℃).

3.1 γ-Al_2O_3 supported solid acid catalysts

Fig.2 shows the XRD patterns of the γ-Al_2O_3 (referred as A) supported solid acid catalysts, and the formation of the tetragonal ZrO_2 phase was observed. The catalyst A prepared by calcining the pseudo-boehmite showed the characteristic peaks at 45.8° and 66.7° corresponding to the diffractions from (400) and (440) planes of γ-Al_2O_3 structure as shown in Fig. 2 (d). The catalysts AS1 and AS2 in Fig.2 (e) and (f) prepared by impregnating γ-Al_2O_3 in sulfuric acid and ammonium persulfate respectively showed a substantially same XRD pattern as catalyst A. For γ-Al_2O_3/ZrO_2, SO_4^{2-}/γ-Al_2O_3/ZrO_2 and $S_2O_8^{2-}$/γ-Al_2O_3/ZrO_2 coded as AZ, AZS1 and

AZS2 respectively, the 2θ peaks at 30.5°, 35.3°, 50.9° and 60.5° were attributed to the diffractions from (101), (110), (112) and (211) planes of tetragonal zirconia [26]. The corresponding diffraction peaks for γ-Al_2O_3 in Fig.2 (a-c) became less detectable than in (d-f) due to the reduction of relative concentration.

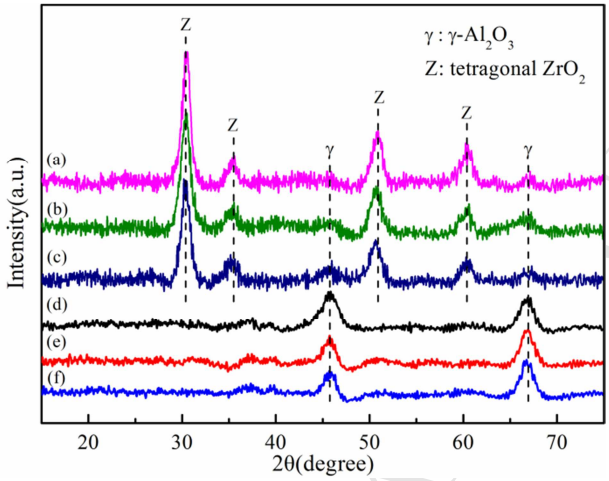


Figure 2. XRD patterns of γ-Al_2O_3 supported solid acid catalysts. (a) AZ, (b) AZS1, (c) AZS2, (d) A, (e) AS1, (f) AS2.

Control experiments showed that the value of Y_{CPL} was only less than 40% in the absence of a catalyst for the MC nylon hydrolysis (Fig.3). The AS1 and AS2 were more active than catalyst A and gave an almost similar catalytic performance, with 87-90% yield of water-soluble products. Compared with catalyst A, AS1, and AS2, the BET surface area of solid catalyst AZ, AZS1 and AZS2 showed a slight decrease (Supplementary, Table S2) due to the larger particle size (Supplementary, Fig.S2) caused by the precipitate of ZrO_2 crystal. However, the yield of monomer CPL was increased obviously especially from 38% to 62% using the catalyst AZS1. That could be explained by the strong acidity of AZS1 and its largest acid amount among the γ-Al_2O_3 supported solid acid catalysts (Supplementary, Fig.S3), which was attributed

to the enhanced acid sites after the incorporation of ZrO_2 and SO_4^{2-} [27].

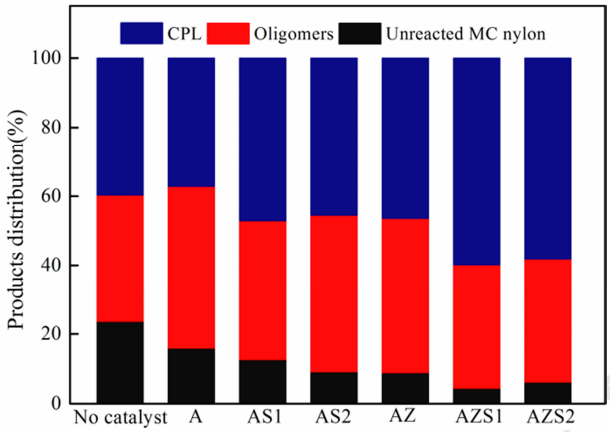


Figure 3. Products distribution of MC nylon hydrolysis using different γ-Al_2O_3 supported solid acid catalysts (Temperature: 345 ℃, Time: 30 min, Catalyst: 0.5 g).

The relationship between the MC nylon hydrolysis effects in subcritical water and the amount of solid acid AZS1 is presented in Fig.4. With the increasing content of solid catalyst, the value of the hydrolysis degree and the yield of CPL which were two important evaluated factors of the degradation, increased obviously when the mass of catalyst was less than 0.5 g. The continuous increasing addition of AZS1 did not lead to the further improvement of the hydrolysis degree or monomer yield.

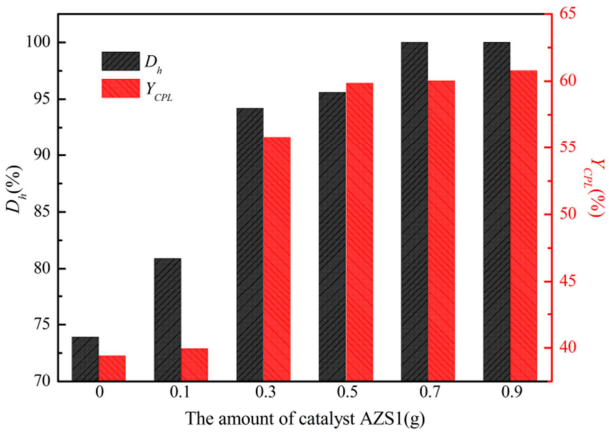


Figure 4. Effect of the catalyst amount on the hydrolysis of MC nylon catalyzed by AZS1 (Temperature: 345 ℃, Time: 30 min).

The catalyst recycling studies using AZS1 (Fig. 5) showed an evident decrease in the yield of CPL after the first recycling. The R_t value also sustained a slight decline in the subsequent cycles, indicating that the recovered catalyst without any treatment was deactivated in the successive runs [28]. The deactivation could result from the organic fouling, which was caused by the adsorbed intermediate oligomer products during the reaction, and the leaching of the sulfonic acid groups [15]. Some masked active sites could be regenerated by calcining the catalyst while the leaching would result in a permanent loss of catalytic activity (Supplementary, Table S2).

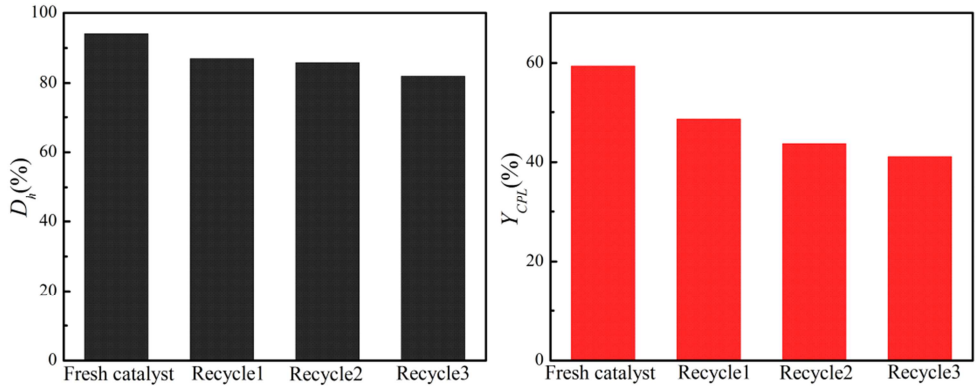


Figure 5. Recycling study of catalyst AZS1 in the hydrolysis of MC nylon (Temperature: 345 ℃, Time: 30 min, Catalyst: 0.5 g).

3.2 H-form zeolites

Since the zeolites are well-established solid acid in the chemical industry providing high surface area, large thermal operating window and good recyclability [29], different H-form zeolites were considered to be applied in the catalyzed hydrolysis reaction of MC nylon (Fig. 6). Compared with the initial blank experiment, the application of H-form zeolites showed a significantly increased yield of monomer CPL, especially the type of Hβ zeolites. Zeolite Hβ-25 gave the highest D_h (96.8%)

together with Y_{CPL} (62.3%) among the employed zeolites. Although the total acid amount of HUSY is larger than Hβ type which is attributed to its lower Si/Al ratio, the most acid sites in Hβ zeolite are strong acid sites (Supplementary, Table S3 and Fig. S4). Besides, the micropore size of Hβ type zeolite is larger than the H-ZSM-5 zeolite, which is facilitated for the transfer of bulk molecules. That is believed to be the two essential factors for accelerating the hydrolysis degradation [30]. The hydrolysis experiments of MC nylon carried out at a lower temperature (320 ℃) using the H-form zeolites also showed a similar result (Supplementary, Fig.S5).

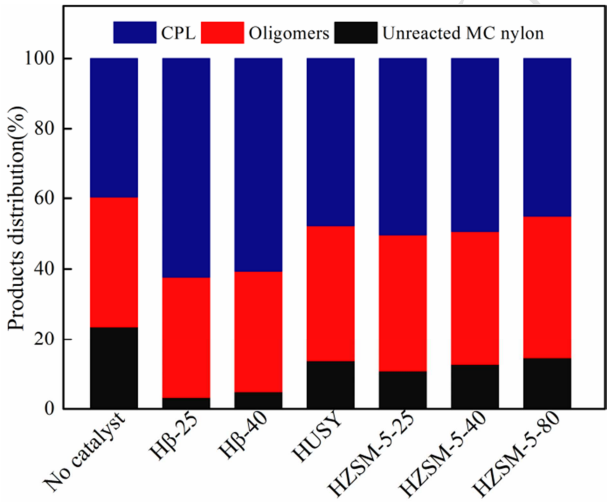


Figure 6. Products distribution of MC nylon hydrolysis using different zeolites catalysts (Temperature: 345 ℃, Time: 30 min, Catalyst: 0.5 g).

When the amount of the zeolite Hβ-25 was larger than 0.3 g, the catalytic hydrolysis effect of MC nylon was essentially unchanged (Supplementary, Fig.S6). The reuse of the Hβ-25 catalyst was examined for four consecutive reaction runs, and the results are shown in Fig. 7. The hydrolysis degree of waste MC nylon remained stable at 95-100% in the recycling experiments, while the yield of CPL was maintaining in the range of 60-63%. Due to the relatively mild reaction condition in

subcritical water for each run (345 ℃, 30 min), the structure of Hβ-25 regenerated after 4th use did not change in comparison to the fresh zeolite except for a decrease of the surface area and the micropore volume (Supplementary, Fig.S7, Table S3). Therefore the catalytic performance of zeolite Hβ-25 was still stable for the degradation of MC nylon after four repeated hydrolysis reactions.

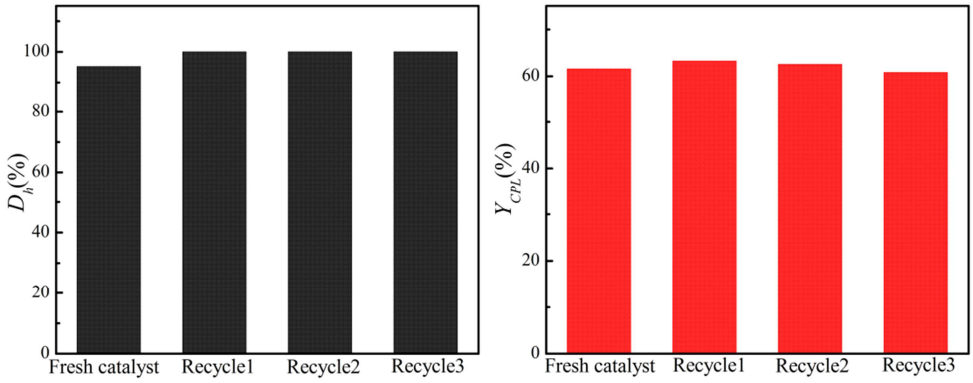
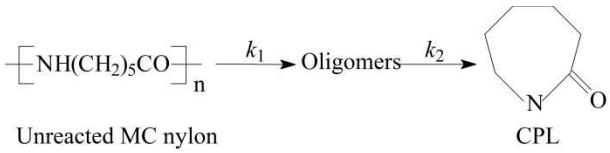


Figure 7. Recycling study of zeolite Hβ-25 in the hydrolysis of MC nylon (Temperature: 345 ℃, Time: 30 min, Catalyst: 0.5 g).

3.3 MC nylon Catalyzed hydrolysis pathway and its kineics

Since the consecutive hydrolysis reaction can be treated as the pseudo-first-order reaction [20, 31], the hydrolysis kinetic model of MC nylon in the subcritical water could be described simply by the following reaction equation (Scheme 2). The solid waste MC nylon first decomposed to oligomers under the hydrothermal condition in Step I, and then the oligomers were hydrolyzed onto the monomer CPL in Step II.



Scheme 2. Main reaction pathway of MC nylon hydrolysis in subcritical water.

The yield rate of oligomers and monomer CPL are given by Eq. (6) and (7).

$$\frac{dC_{Oligomers}}{dt} = k_1 C_{MC\ nylon} - k_2 C_{CPL} \tag{6}$$

$$\frac{dC_{CPL}}{dt} = k_2 C_{Oligomers} \tag{7}$$

When $t = 0$, $C_{MC\ nylon} = C_{MC\ nylon,0}$ and $C_{Oligomers} = C_{CPL} = 0$, and then the concentrations of CPL could be expressed by Eq. (8). The values of k_1 and k_2 were determined by fitting the experimental data with the least squares method.

$$C_{CPL} = C_{MC\ nylon,0}(1 - e^{-k_1 t}) - C_{MC\ nylon,0}\frac{k_1}{k_1 - k_2}(e^{-k_2 t} - e^{-k_1 t}) \tag{8}$$

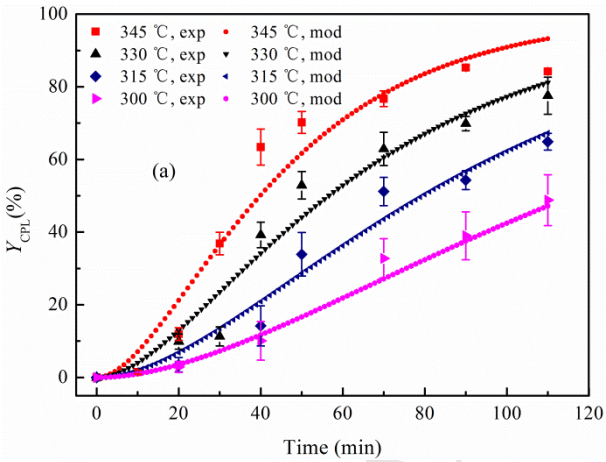
Because MC nylon is solid, its content can be expressed in terms of weight. At the same time C_{CPL} can be replaced as Y_{CPL}.The temperature dependence of the rate constants can be represented by Arrhenius equation which is shown in Eq.(9):

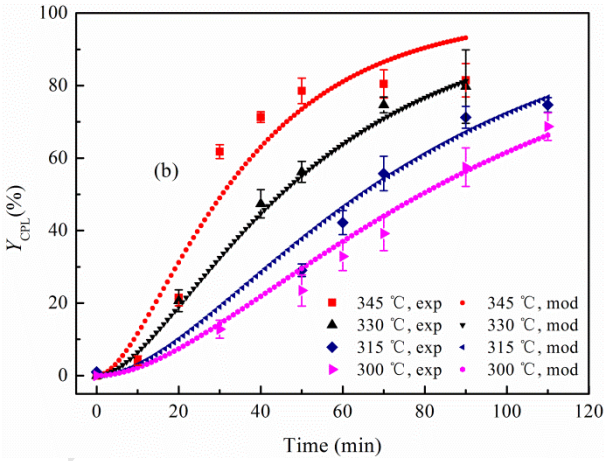
$$k = Ae^{-\frac{E}{RT}} \tag{9}$$

where E is the apparent activation energy.

The comparison between the experimental and the kinetic fitting values of Y_{CPL} for the non-catalyzed hydrolysis at different temperatures is shown in Figure 8(a), while for catalyzed hydrolysis reaction using catalysts AZS1 and Hβ-25 are shown in Figure 8(b) and (c), respectively. The corresponding kinetic parameters for MC nylon hydrolysis were fitted by kinetic Eq.8, evaluated by the non-linear least squares regression method and showed in Table 1. Once the oligomer generated in the hydrolysis system, the yield of monomer CPL began to increase with the increasing reaction time. For the hydrolysis reaction carried on from 300 ℃ to 345 ℃, the value of k_1 was larger than k_2, which indicated that the conversion of oligomers onto the CPL is the critical step during the whole hydrolysis of MC nylon. In the time

range of 70-120 min at a higher temperature (345 ℃), the experimental values were observed deviating from the model curve. We believe that the decrement of monomer yield was caused by the reversed polymerization as opposing reaction of the decomposition.





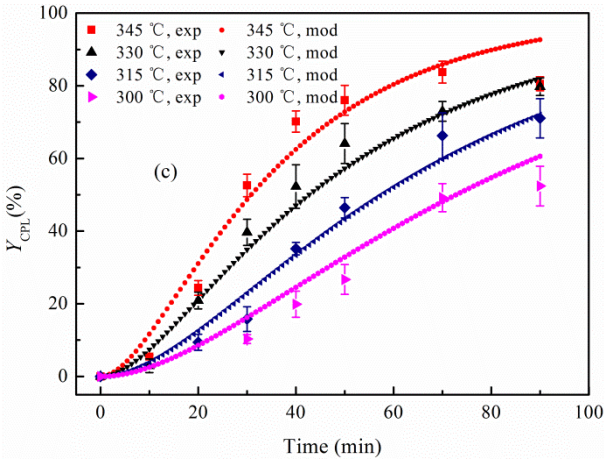


Figure 8. Comparison between the experimental (Mean ± SD) and model values of Y_{CPL} vs. hydrolysis time at different temperatures (a) Hydrolysis non-catalyzed; (b) Hydrolysis catalyzed by AZS1; (c) Hydrolysis catalyzed by Hβ-25 (Catalyst: 0.3 g).

Table 1 Determined Values for the Kinetic Parameters of MC nylon catalyzed hydrolysis (Catalyst: 0.3 g).

Rate constants(min^{-1})	300 ℃		315 ℃		330 ℃		345 ℃		Ea_1 (kJ)	Ea_2 (kJ)
	k_1	k_2	k_1	k_2	k_1	k_2	k_1	k_2		
Hydrolysis non-catalyzed	0.019	0.011	0.025	0.018	0.054	0.019	0.063	0.030	84.21	59.87
Hydrolysis catalyzed by AZS1	0.037	0.014	0.039	0.019	0.081	0.022	0.11	0.034	76.06	50.50
Hydrolysis catalyzed by Hβ-25	0.040	0.015	0.051	0.019	0.10	0.022	0.11	0.033	73.80	47.87

The formation of oligomers contained the melting of solid MC nylon and the chain scission reaction, which resulted in the higher activation energy for Step I (Table 1). Therefore, the first step would be more vulnerable to the effects of temperature. For catalyzed hydrolysis of MC nylon, the increasing temperature and reaction time showed similar improvement as non-catalyzed hydrolysis. Both the catalyst AZS1 and zeolite Hβ-25 presented an obvious catalytic activity in comparison with non-catalyzed reactions. Due to the lack of efficient contact between the solid/molten MC nylon and the acid sites of the solid catalyst in Step I, the solid catalysts played a more important role in the second step, which led to a larger decrement of Ea_2 value.

As the degradation rate was accelerated more obviously by Hβ-25, the specific pathway was studied by examining the compositional changes of liquid phase products during the catalyzed decomposition. The water-soluble cyclic and linear oligomers were referred as Cn and Ln respectively, where n represented the number of

-$(CH_2)_5$- segment in the molecule. The mass spectrum of catalyzed hydrolytic liquid and the corresponding identified degradation products determined by LC-MS are given in Fig.S8 and Table S4. Meanwhile, the specific chromatogram of the oligomers at different time (345 ℃) obtained from the catalyzed hydrolysis reactions are shown in Fig.S9 - Fig.S12. The relative peak area of the compositions, which could directly reflect the comparison of the concentration changing, are calculated and shown in Fig.9 and Fig.10 for catalyzed hydrolysis using AZS1 and Hβ-25 respectively.

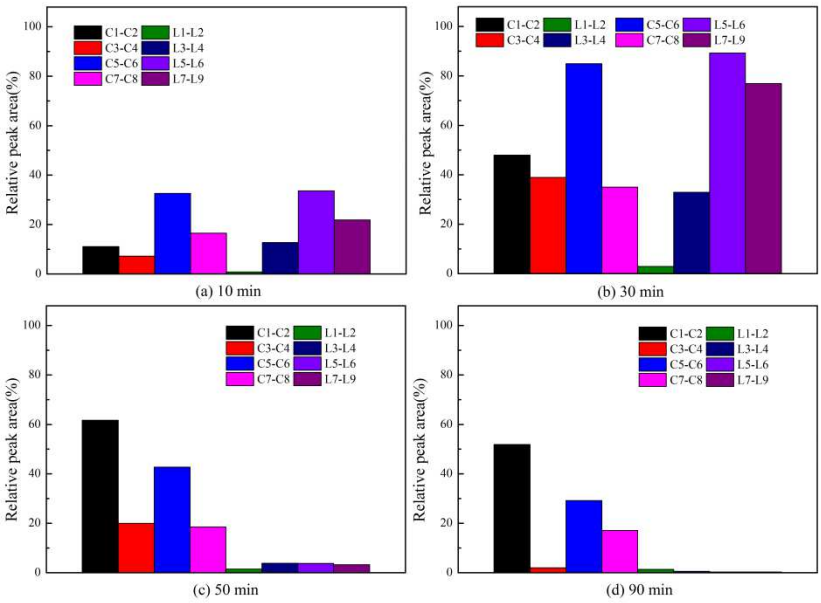


Figure 9. The relative peak area of the compositions obtained during the catalyzed hydrolysis of MC nylon using catalyst AZS1 (Temperature: 345 ℃, Catalyst: 0.3 g).

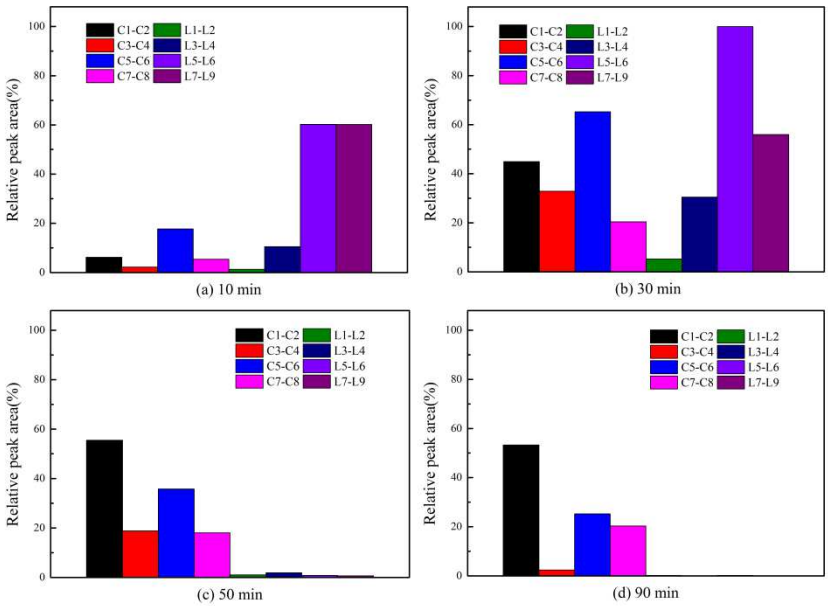


Figure 10. The relative peak area of the compositions obtained during the catalyzed hydrolysis of MC nylon using zeolite Hβ-25 (Temperature: 345 ℃, Catalyst: 0.3 g).

The concentration of the components with n≥3 increased in the first 30 min and then decreased rapidly, which indicated that the oligomers were the intermediates in the hydrolysis. It is consistent with the hydrolysis mechanism and the kinetic model. What worth mentioned was that at the end of the decomposition, the linear oligomers were reacted exhaustively while the concentration of some cyclic oligomers with n>5 decreased slowly. This was caused by some oligomerization. The reaction pathway using different solid acids showed similar trend, except that the addition of zeolite Hβ-25 preferred to accelerate the generation and consumption of the linear intermediate oligomers during the process. This phenomenon could be explained by the difference of catalyst surface structure. When the water-soluble oligomers were produced after the cleavage of the amide bond, the linear intermediates could enter the micropores of zeolite and react on the acid sites of the internal surface to form shorter linear oligomers. Due to the absence of microporous structure which would obstruct

the entering of cyclic oligomers with the large diameter, the catalyst AZS1 would accelerate the production of both linear and cyclic oligomers.

4. Conclusion

The waste MC nylon was readily hydrolyzed to the polymeric monomer ε-caprolactam (CPL) in the presence of the solid acid catalyst. The results revealed that the hydrolysis rate and the reaction pathway depended on the type of solid acids. Although the solid acid catalysts $SO_4^{2-}/\gamma\text{-}Al_2O_3/ZrO_2$ (AZS1) acted as an effective catalyst for accelerating the hydrolysis reaction, showed a loss of activity after the first reaction run due to the leaching of the sulfonic acid groups in hot water. The zeolite Hβ-25 afforded the best results in terms of both activity and catalyst stability among the solid acids used in this work. During the catalytic degradation of waste MC nylon in water, the linear oligomers preferred to be generated and then consumed quickly when using Hβ-25, because the microporous structure obstructed the cyclic intermediate oligomers entering. The combination of the purely green solvent water and an effective solid catalyst with high potential for improving the hydrolysis would make the degradation of MC nylon and the recycling of economic monomer CPL worthy of further exploration.

Acknowledgement

The authors gratefully acknowledge financial support from the National Natural Science Foundation of China (No. 21174034).

References

[1] Fuentes-Ordóñez EG, Salbidegoitia JA, González-Marcos MP, González-Velasco JR. Mechanism and kinetics in catalytic hydrocracking of polystyrene in solution. Polym Degrad Stab. 2016;124:51-9.

[2] Su K-H, Lin J-H, Lin C-C. Influence of reprocessing on the mechanical properties and structure of polyamide 6. J Mater Process Technol. 2007;192-193:532-8.

[3] Kamimura A, Oishi Y, Kaiso K, Sugimoto T, Kashiwagi K. Supercritical secondary alcohols as useful media to convert polyamide into monomeric lactams. ChemSusChem. 2008;1:82-4.

[4] Bozi J, Blazsó M. Catalytic modification of pyrolysis products of nitrogen-containing polymers over Y zeolites. Green Chem. 2009;11:1638-45.

[5] Medina-Ramos W, Mojica MA, Cope ED, Hart RJ, Pollet P, Eckert CA, et al. Water at elevated temperatures (WET): reactant, catalyst, and solvent in the selective hydrolysis of protecting groups. Green Chem. 2014;16:2147-55.

[6] Bozi J, Mihályi MR, Blazsó M. Study on temperature dependence of catalytic thermal decomposition of polyamides and polyurethanes mixed with acidic zeolites. J Anal Appl Pyrolysis. 2013;101:103-10.

[7] Wang W, Meng L, Huang Y. Hydrolytic degradation of monomer casting nylon in subcritical water. Polym Degrad Stab. 2014;110:312-7.

[8] Fan J, De bruyn M, Budarin VL, Gronnow MJ, Shuttleworth PS, Breeden S, et al. Direct microwave-assisted hydrothermal depolymerization of cellulose. J Am Chem Soc. 2013;135:11728-31.

[9] Shukla SR, Harad AM, Mahato D. Depolymerization of nylon 6 waste fibers. J Appl Polym Sci. 2006;100:186-90.

[10] He Y, Hoff TC, Emdadi L, Wu Y, Bouraima J, Liu D. Catalytic consequences of micropore topology, mesoporosity, and acidity on the hydrolysis of sucrose over zeolite catalysts. Catal Sci Technol. 2014;4:3064-73.

[11] Tian J, Wang J, Zhao S, Jiang C, Zhang X, Wang X. Hydrolysis of cellulose by the heteropoly acid $H_3PW_{12}O_{40}$. Cellulose. 2010;17:587-94.

[12] Qian X, Lei J, Wickramasinghe SR. Novel polymeric solid acid catalysts for cellulose hydrolysis. RSC Adv. 2013;3:24280.

[13] Qi X, Lian Y, Yan L, Smith RL. One-step preparation of carbonaceous solid acid catalysts by hydrothermal carbonization of glucose for cellulose hydrolysis. Catal Commun. 2014;57:50-4.

[14] Chakinala AG, van Swaaij WPM, Kersten SRA, de Vlieger D, Seshan K, Brilman DWF. Catalytic Reforming of Glycerol in Supercritical Water over Bimetallic Pt–Ni Catalyst. Ind Eng Chem Res. 2013;52:5302-12.

[15] Carà PD, Pagliaro M, Elmekawy A, Brown DR, Verschuren P, Shiju NR, et al. Hemicellulose hydrolysis catalysed by solid acids. Catal Sci Technol. 2013;3:2057-61.

[16] Lv B, Zhao G, Li D, Liang C. Dechlorination and oxidation for waste poly(vinylidene chloride) by hydrothermal catalytic oxidation on Pd/AC catalyst. Polym Degrad Stab. 2009;94:1047-52.

[17] Faba L, Kusema BT, Murzina EV, Tokarev A, Kumar N, Smeds A, et al. Hemicellulose hydrolysis and hydrolytic hydrogenation over proton- and metal modified beta zeolites. Micropor Mesopor Mat. 2014;189:189-99.

[18] Zhou L, Shi M, Cai Q, Wu L, Hu X, Yang X, et al. Hydrolysis of hemicellulose catalyzed by hierarchical H-USY zeolites – The role of acidity and pore structure. Micropor Mesopor Mat. 2013;169:54-9.

[19] Watanabe M, Inomata H, Arai K. Catalytic hydrogen generation from biomass (glucose and cellulose) with ZrO_2 in supercritical water. Biomass Bioenerg. 2002;22:405-10.

[20] Li X-K, Lu H, Guo W-Z, Cao G-P, Liu H-L, Shi Y-H. Reaction kinetics and mechanism of catalyzed hydrolysis of waste PET using solid acid catalyst in supercritical CO_2. AICHE J. 2015;61:200-14.

[21] Yanagihara N, Ohgane K. Studies on the oxidative degradation of nylons by nitrogen dioxide in supercritical carbon dioxide. Polym Degrad Stab. 2013;98:2735-41.

[22] Liu L, Tang LH, Wu YQ, Ni YH, Zhu ZB. Degradation process investigation of thermoplastic polyurethane elastomer in supercritical methanol. Polym Degrad Stab. 2013;98:2520-8.

[23] Cantero DA, Bermejo MD, Cocero MJ. Governing chemistry of cellulose hydrolysis in supercritical water. ChemSusChem. 2015;8:1026-33.

[24] Iwaya T, Sasaki M, Goto M. Kinetic analysis for hydrothermal depolymerization of nylon 6. Polym Degrad Stab. 2006;91:1989-95.

[25] Xu S, Zhao X, Ye L. Effect of Heat Treatment on the Structure and Properties of MC Nylon 6. Polym-Plast Technol Eng. 2012;51:689-95.

[26] Liu E, Locke AJ, Frost RL, Martens WN. Sulfated fibrous ZrO_2/Al_2O_3 core and shell nanocomposites: A novel strong acid catalyst with hierarchically macro–mesoporous nanostructure. J Mol Catal A: Chem. 2012;353-354:95-105.

[27] Reddy BM, Sreekanth PM, Yamada Y, Kobayashi T. Surface characterization and catalytic activity of sulfate-, molybdate- and tungstate-promoted Al_2O_3–ZrO_2 solid acid catalysts. J Mol Catal A: Chem. 2005;227:81-9.

[28] Peng L, Lin L, Zhang J, Shi J, Liu S. Solid acid catalyzed glucose conversion to ethyl levulinate. Applied Catalysis A: General. 2011;397:259-65.

[29] Saravanamurugan S, Riisager A. Zeolite-catalyzed isomerization of tetroses in aqueous medium. Catal Sci Technol. 2014;4:3186-90.

[30] Zhou L, Liu Z, Bai Y, Lu T, Yang X, Xu J. Hydrolysis of cellobiose catalyzed by zeolites—the role of acidity and micropore structure. J. Energ. Chem. 2016;25:141-5.

[31] Hocker S, Rhudy AK, Ginsburg G, Kranbuehl DE. Polyamide hydrolysis accelerated by small weak organic acids. Polymer. 2014;55:5057-64.

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